

Devialet RS-232 Control

Installation and Configuration Guide

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Installing Devialet-rs232

1 Introduction

This document is will guide you to al the step to install and configure all components to control your Devialet Expert Pro 220 and 440 via rs-232.

2 Prerequisites

One older Raspberry Pi is sufficient. I started with the 1B, now use the Zero and use the 1B for testing.

2.1 Raspberry Pi 1B

- SD card (minimal 8GB)
- Wired UTP Connection between Raspberry Pi 1B and Home Assistant (same subnet e.g VLAN)

2.2 Raspberry Pi Zero 2 W

- Pi Zero 2 W
- micro SD card (minimal 8GB)
- WiFi Connection between Raspberry Pi Zero 2 W and Home Assistant (same subnet e.g VLAN)

2.3 USB - Serial

Pi Zero 2 W: Micro USB – USB-A OTG cable

USB A to Serial converter with FTDI chip

- USB-A (m) naar 9-pins SUB-D met moeren (v) seriële RS232 adapter / FTDI chip - 1,8 meter



Figure 1. USB A to Serial converter with FTDI chip

Serial to 3,5mm converter (based on pin-laout from Devialet manual)

- StarTech.com 1m DB9 naar 3.5mm Seriële Kabel for Serial Device Configuratie, RS232 DB9 Male

3,5 mm TRS	DB9 RS-232
Tip	Pin 3 (TX)
Ring	Pin 2 (RX)
Sleeve	Pin 5 (GND)

Figure 2. Serial to 3,5mm converter pinout

<https://www.bol.com/nl/nl/p/startech-com-cavo-seriale-db9-a-jack-3-5-mm-da-1m-adattatore-seriale-rs232-9-pin-a-jack-3-5-m-m-per-la-configurazione-e-calibrazione-di-dispositivi-seriali-proiettori-digital-signage-e-tv-tramite-jack-audio/9300000129874558/>



Figure 3. Serial to 3,5mm converter

2.4 Software

- Raspberry Pi Imager v2.0.10 for macOS (to install a fresh Linux image on the raspberry). Download installation from <https://www.raspberrypi.com/software/>
- Terminal for macOS (to use SSH for installing components and configure the raspberry)
- Home Assistant (optional) if you want to control the Devialet from Home Assistant via a custom dashboard or automation (to Power On/Off, set Source etc.)

2.5 Devialet documentation

D-Premier-RS232.pdf: <https://www.deviallet.com/en-us/docs/D-Premier-RS232.pdf>

3 Prepare the Devialet

Use the Devialet configurator to enable RS232 Control

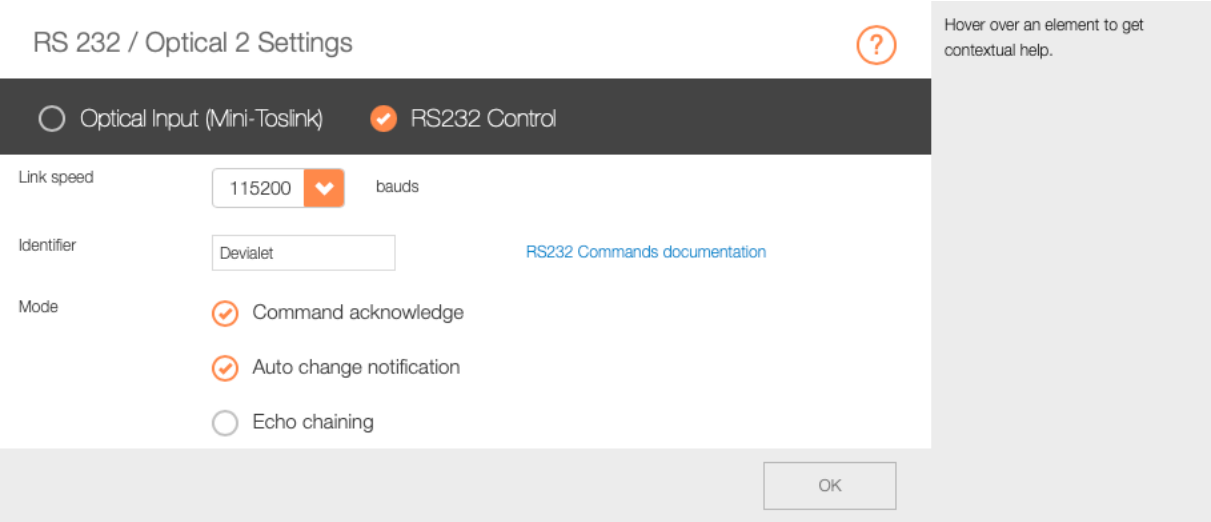


Figure 4. RS 232 / Optical 2 Settings

The 3,5mm RS232 plug fits in the Optical 2 input:

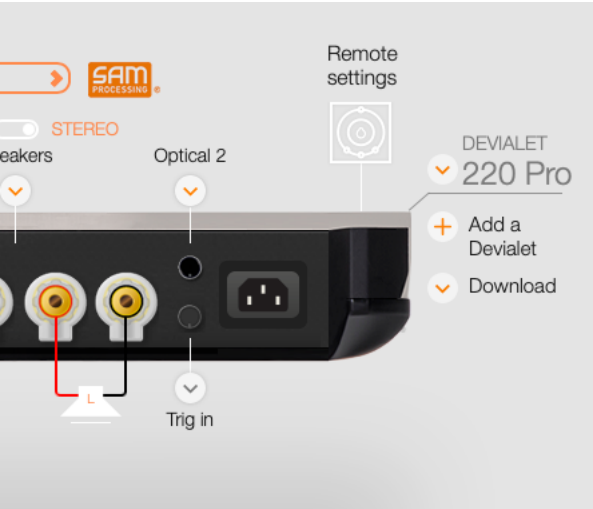


Figure 5. Optical 2 port

4 Prepare the Raspberry Pi

Preparation of the SD-card is similar for Raspberry Pi 1B and Pi Zero 2 W.

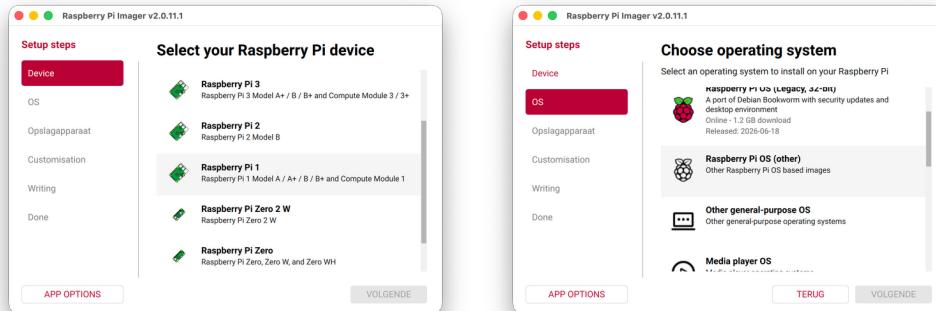


Figure 6. Raspberry Pi imager

1. Choose Raspberry Pi 1
2. Choose Raspberry Pi OS (other)

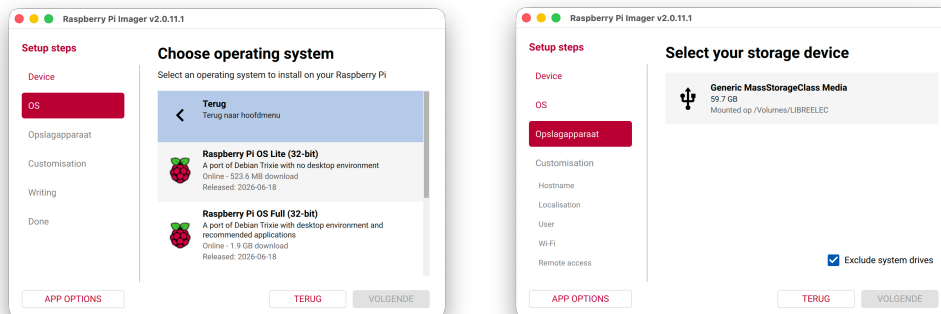


Figure 7. Raspberry Pi imager

3. Choose Raspberry Pi OS Lite (32-bit)
4. Select SD Card

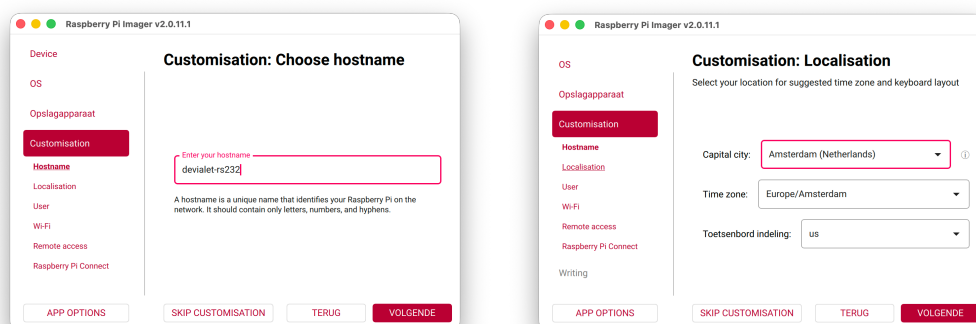


Figure 8. Raspberry Pi imager

5. Enter your hostname: devialat-rs232
6. Fill in Localisation

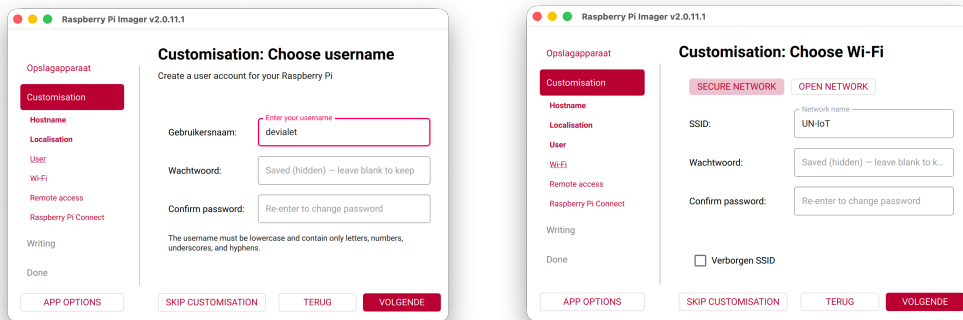


Figure 9. Raspberry Pi imager

7. Choose Username: devialet
8. Choose Password: devialet
9. Choose Wi-Fi (Although the Pi 1B has no onboard WiFi module)

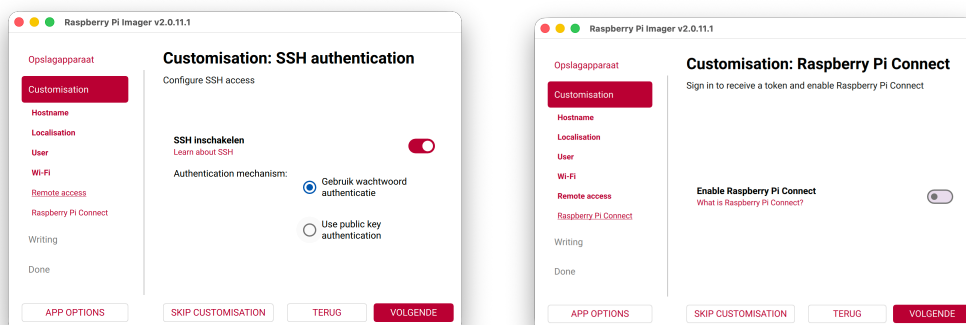


Figure 10. Raspberry Pi imager

10. Enable SSH
11. Do not Enable Raspberry Pi Connect

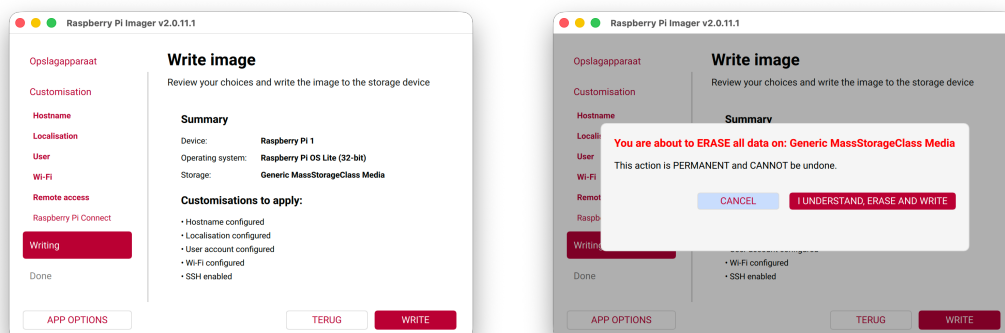


Figure 11. Raspberry Pi imager

12. Write the image to the SD Card
13. Put the SD Card in the Raspberry Pi (connected to the network)
14. Boot the Raspberry Pi by plugging in the power adapter

5 Complete the first-boot and setup

Start Terminal (Mac OSX) or similar SSH client for Windows (like Putty)

Only the **blue texts** have to be entered at the prompt, the rest is output from the Pi.

5.1 First login with SSH

```
eddy@MacBookPro ~ % ssh devialet@devialet-rs232.local
The authenticity of host 'devialet-rs232.local (fe80::ba27:ebff:fe35:6c3%en0)'
can't be established.
ED25519 key fingerprint is: SHA256:s7uPTlBktI5z66RUh2emzPtrRkXIaCg8YpeK8/O/NQo
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'devialet-rs232.local' (ED25519) to the list of
known hosts.
devialet@devialet-rs232.local's password: *****
Linux devialet-rs232 6.18.34+rpt-rpi-v6 #1 Raspbian 1:6.18.34-1+rpt1 (2026-06-
09) armv6l
```

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

5.2 Refresh local package list

```
devialet@devialet-rs232:~ $ sudo apt update
[sudo] password for devialet: *****
Hit:1 http://archive.raspberrypi.com/debian trixie InRelease
Hit:2 http://raspbian.raspberrypi.com/raspbian trixie InRelease
104 packages can be upgraded. Run 'apt list --upgradable' to see them
```

5.3 Install Minicom

```
devialet@devialet-rs232:~ $ sudo apt install minicom
Installing:
  minicom
```

```
Installing dependencies:
  lrzsz
```

```
Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 104
  Download size: 313 kB
  Space needed: 1,268 kB / 57.7 GB available
```

```
Continue? [Y/n] y
Get:1 http://mirror.transip.net/raspbian/raspbian trixie/main armhf lrzsz
armhf 0.12.21rc-0.2 [76.0 kB]
Get:2 http://mirror.netzwerke.de/raspbian/raspbian trixie/main armhf minicom
armhf 2.10-1 [237 kB]
Fetched 313 kB in 1s (315 kB/s)
Selecting previously unselected package lrzsz.
(Reading database ... 67323 files and directories currently installed.)
Preparing to unpack .../lrzsz_0.12.21rc-0.2_armhf.deb ...
Unpacking lrzsz (0.12.21rc-0.2) ...
Selecting previously unselected package minicom.
Preparing to unpack .../minicom_2.10-1_armhf.deb ...
Unpacking minicom (2.10-1) ...
Setting up minicom (2.10-1) ...
Setting up lrzsz (0.12.21rc-0.2) ...
```

Processing triggers for man-db (2.13.1-1) ...

5.4 Verify and Setup USB-RS232 adapter

```
devialet@devialet-rs232:~ $ lsusb
```

```
ls -l /dev/ttyUSB*
```

```
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

```
Bus 001 Device 002: ID 0424:9512 Microchip Technology, Inc. (formerly SMSC)  
SMC9512/9514 USB Hub
```

```
Bus 001 Device 003: ID 0424:ec00 Microchip Technology, Inc. (formerly SMSC)  
SMSC9512/9514 Fast Ethernet Adapter
```

```
Bus 001 Device 004: ID 0403:6001 Future Technology Devices International, Ltd  
FT232 Serial (UART) IC
```

```
crw-rw---- 1 root dialout 188, 0 Jun 18 02:17 /dev/ttyUSB0
```

Setup Minicom (set defaults on 115200 N 8 1)

```
devialet@devialet-rs232:~ $ sudo minicom -s
```

Serial port setup:



Figure 12. Minicom Serial port setup

A – Serial Device: `/dev/ttyUSB0` (identical to previous)

E – BPS/Par/Bits: 115200 8N1 (identical to the Devialet Setup in step 3)

F – L : No

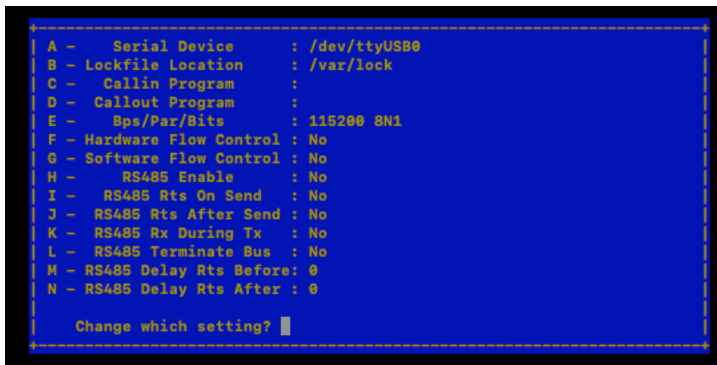


Figure 13. Minicom Serial port setup

Save setup as default (dfl):

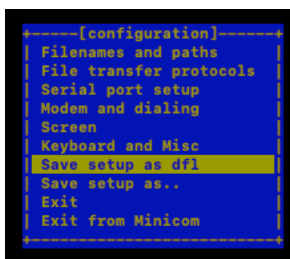


Figure 14. Minicom Serial port setup

Exit from Minicom

5.5 Startup Minicom

devialet@devialet-rs232:~ \$ minicom

```
Welcome to minicom 2.10
OPTIONS: I18n
Port /dev/ttyUSB0 [U]
Press CTRL-A Z for help on special keys
```

Figure 15. Minicom

CTRL-AZ

```
Welcome to minicom 2.10
OPTIONS: I18n
Port /dev/ttyUSB0 [U]
Press CTRL-A Z for help on sp
```

Minicom Command Summary

Commands can be called by CTRL-A <key>

Main Functions		Other Functions
Dialing directory...D	run script (Go)....G	Clear Screen.....C
Send files.....S	Receive files.....R	cOnfigure Minicom..O
comm Parameters....P	Add linefeed.....A	Suspend minicom....J
Capture on/off....L	Hangup.....H	eXit and reset....X
send break.....F	initialize Modem...M	Quit with no reset.Q
Terminal settings..T	run Kermit.....K	Cursor key mode...I
lineWrap on/off...W	local Echo on/off..E	Help screen.....Z
Paste file.....Y	Timestamp toggle...N	scroll Back.....B
Add Carriage Ret...U		

Select function or press Enter for none.

Figure 16. Minicom Command Summary

“E” for local Echo on

6 Verify Devialet RS-232 communication

6.2 Power On the Devialet and watch Minicom

Power On the Devialet 220 or 440 (from stand-by) with the normal Devialet remote or iOS App, and you will see something similar in Minicom:

```
Welcome to minicom 2.10
OPTIONS: I18n
Port /dev/ttyUSB0 [U]
Press CTRL-A Z for help on special keys

[Devialet>START:1]
[Devialet>SOURCE:Room Ready]
[Devialet>BALANCE:0]
[Devialet>TONE_CONTROL:0]
[Devialet>BASS:0.0]
[Devialet>TONE_CONTROL:0]
[Devialet>TREBLE:0.0]
[Devialet>MATCHING_LEVEL:0]
[Devialet>POWER:0]
[Devialet>PHASE:0]
[Devialet>VOLUME:-40.0]
[Devialet>MUTE:0]
[Devialet>SUBSONIC_FILTER:0]
[Devialet>SUBWOOFER:1]
[Devialet>CURVE:4]
[Devialet>PREOUT:1]
[Devialet>USER_DSP:0]
[Devialet>VOLUME_BYPASS:0]
[Devialet>POWER:1]
[Devialet>START:0]
```

Figure 17. Devialet RS-232 communication

Copy past this commands as test:

```
[Devialet>VOLUME=?][Devialet>VOLUME:-40.0]
[Devialet>VOLUME=100][Devialet>INVALID PARAMETER]
[Devialet>VOLUME=-70][Devialet>INVALID COMMAND]
[Devialet>VOLUME=-20][Devialet>VOLUME=ACK]
[Devialet>VOLUME:-20.0]
```

Figure 18. Devialet commands

[Devialet>VOLUME=?] **only returns the current value of volume**

[Devialet>VOLUME=100] If you entered an invalid value, you will see **"INVALID PARAMETER"**

[Devialet>VOLUME=70] This is an **INVALID COMMAND** (mistype volume)

[Devialet>VOLUME=-20] returns an **ACK** = acknowledge, and after that the new value returned

If this all works as described above, the basic setup is successfully completed. What is next is the installation of the modules to run the remote control software.

6.2 Close Minicom

CTRL + A Z X



Figure 19. Leave Minicom

7 Useful commands from SSH

7.1 Shutdown Raspberry Pi

```
devialet@devialet-rs232:~ $ sudo poweroff
[sudo] password for devialet:*****
devialet@devialet-rs232:~ $ Connection to devialet-rs232.local closed by
remote host.
Connection to devialet-rs232.local closed.
client_loop: send disconnect: Broken pipe
```

Wait for 30 sec....

7.2 Logoff SSH

```
devialet@devialet-rs232:~ $ exit

logout

Connection to devialet-rs232.local closed.
```

7.3 Testing rs232 serial communications

```
devialet@devialet-rs232:~ $ minicom
```

7.4 Test service active

```
devialet@devialet-rs232:~ $ systemctl is-active devialet-rs232.service
active
```

8 Install the devialet-rs232 control software

8.1 How to

Install the devialet software from Github onto the Pi.

8.2 Install modules

```
devialet@devialet-rs232:~ $ sudo apt install python3-serial python3-paho-mqtt  
-y
```

```
[sudo] password for devialet: *****
```

Installing:

```
python3-paho-mqtt python3-serial
```

Suggested packages:

```
python3-wxgtk3.0 | python3-wxgtk
```

Summary:

```
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 104
```

```
Download size: 152 kB
```

```
Space needed: 800 kB / 57.7 GB available
```

```
Get:2 http://mirror.transip.net/raspbian/raspbian trixie/main armhf python3-serial all 3.5-2 [88.2 kB]
```

```
Get:1 http://mirror.netzwerke.de/raspbian/raspbian trixie/main armhf python3-paho-mqtt all 2.1.0-1 [63.4 kB]
```

```
Fetched 152 kB in 1s (191 kB/s)
```

```
Selecting previously unselected package python3-paho-mqtt.
```

```
(Reading database ... 67382 files and directories currently installed.)
```

```
Preparing to unpack .../python3-paho-mqtt_2.1.0-1_all.deb ...
```

```
Unpacking python3-paho-mqtt (2.1.0-1) ...
```

```
Selecting previously unselected package python3-serial.
```

```
Preparing to unpack .../python3-serial_3.5-2_all.deb ...
```

```
Unpacking python3-serial (3.5-2) ...
```

```
Setting up python3-serial (3.5-2) ...
```

```
Setting up python3-paho-mqtt (2.1.0-1) ...
```

8.3 Download installation from Github

```
devialet@devialet-rs232:~ $ wget https://github.com/EddyZure/devialet-rs232/raw/main/devialet-github-20260902-163438.tar.gz
```

```
--2026-09-11 19:43:28-- https://github.com/EddyZure/devialet-rs232/raw/main/devialet-github-20260902-163438.tar.gz
```

```
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)|140.82.121.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/EddyZure/devialet-
rs232/main/devialet-github-20260902-163438.tar.gz [following]
--2026-09-11 19:43:29-- https://raw.githubusercontent.com/EddyZure/devialet-
rs232/main/devialet-github-20260902-163438.tar.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.111.133, 185.199.110.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 600843 (587K) [application/octet-stream]
Saving to: 'devialet-github-20260902-163438.tar.gz'
```

```
devialet-github-20260902-163438
100%[=====>] 586.76K
1.46MB/s in 0.4s
```

```
2026-09-11 19:43:29 (1.46 MB/s) - 'devialet-github-20260902-163438.tar.gz'
saved [600843/600843]
```

8.4 Unpack downloaded archive

```
devialet@devialet-rs232:~ $ tar -xzf devialet-github-20260902-163438.tar.gz
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.lastuseddate#PS'
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.lastuseddate#PS'
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.lastuseddate#PS'
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.FinderInfo'
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.quarantine'
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.lastuseddate#PS'
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.quarantine'
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.lastuseddate#PS'
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'
tar: Ignoring unknown extended header keyword
'LIBARCHIVE.xattr.com.apple.FinderInfo'
```

Note: the errors above are related to my Apple Macbook in this first version and have no effect!

8.5 Verify download and unpacked files

```
devialet@devialet-rs232:~ $ ls -la
```

```
total 624
drwx----- 5 devialet devialet  4096 Sep 11 19:43 .
drwxr-xr-x  3 root      root      4096 Jun 18 02:18 ..
-rw-r--r--  1 devialet devialet   220 Jun 18 02:07 .bash_logout
-rw-r--r--  1 devialet devialet  3523 Jun 18 02:07 .bashrc
drwxr-xr-x  3 devialet devialet  4096 Sep 11 16:28 .config
-rw-rw-r--  1 devialet devialet 600843 Sep 11 19:43 devialet-github-20260902-163438.tar.gz
drwxr-xr-x  6 devialet devialet  4096 Sep  2 16:34 devialet-pi
-rw-r--r--  1 devialet devialet   807 Jun 18 02:07 .profile
drwx----- 2 devialet devialet  4096 Jun 18 02:19 .ssh
-rw-r--r--  1 devialet devialet     0 Sep 11 17:02 .sudo_as_admin_successful
-rw-rw-r--  1 devialet devialet   215 Sep 11 19:43 .wget-hsts
```

8.6 List devialet-pi files

```
devialet@devialet-rs232:~ $ cd ~/devialet-pi
```

```
find . -maxdepth 3 -type f -print | sort
```

```
./docs/._.DS_Store
./docs/.DS_Store
./docs/images/.gitkeep
./docs/images/iphone-home-assistant.jpg
./docs/images/iphone-web-interface.jpg
./env.example
./gitignore
./._README.md
./README.md
./._requirements.txt
./requirements.txt
./scripts/backup.sh
./scripts/check-secrets.sh
./scripts/install.sh
./scripts/restore.sh
./SECURITY.md
./src/devialet_bridge.py
./src/devialet_web_assets/._header-power-off.jpg
./src/devialet_web_assets/header-power-off.jpg
```

```
./src/devialet_web_assets/._header-power-on.jpg
./src/devialet_web_assets/header-power-on.jpg
./src/devialet_web_assets/sam-processing-logo.png
./src/._.DS_Store
./src/.DS_Store
./systemd/devialet-bridge.service
./._VERSION
./VERSION
```

8.7 Run install script

```
devialet@devialet-rs232:~/devialet-pi $ sudo ./scripts/install.sh
[sudo] password for devialet: *****
Hit:1 http://archive.raspberrypi.com/debian trixie InRelease
Get:2 http://raspbian.raspberrypi.com/raspbian trixie InRelease [15.0 kB]
Get:3 http://raspbian.raspberrypi.com/raspbian trixie/main armhf Packages
[15.7 MB]
Fetched 15.7 MB in 29s (546 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.13.5-1).
python3 set to manually installed.
python3-venv is already the newest version (3.13.5-1).
The following additional packages will be installed:
  javascript-common libexpat1 libexpat1-dev libjs-jquery libjs-sphinxdoc
  libjs-underscore libpython3-dev libpython3.13
  libpython3.13-dev libpython3.13-minimal libpython3.13-stdlib python3-dev
  python3-packaging python3-wheel python3.13
  python3.13-dev python3.13-minimal python3.13-venv zlib1g-dev
Suggested packages:
  apache2 | lighttpd | httpd python3-setuptools
The following NEW packages will be installed:
  javascript-common libexpat1-dev libjs-jquery libjs-sphinxdoc libjs-
  underscore libpython3-dev libpython3.13-dev python3-dev
  python3-packaging python3-pip python3-wheel python3.13-dev zlib1g-dev
The following packages will be upgraded:
  libexpat1 libpython3.13 libpython3.13-minimal libpython3.13-stdlib
  python3.13 python3.13-minimal python3.13-venv
7 upgraded, 13 newly installed, 0 to remove and 97 not upgraded.
Need to get 15.4 MB of archives.
```

After this operation, 33.1 MB of additional disk space will be used.

Get:1 <http://archive.raspberrypi.com/debian> trixie/main armhf python3-pip all 25.1.1+dfsg-1+rpt1 [2,266 kB]

Get:2 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf libexpat1 armhf 2.8.3-1~deb13u1 [101 kB]

Get:3 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf python3.13-venv armhf 3.13.5-2+deb13u3 [5,480 B]

Get:4 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf libpython3.13 armhf 3.13.5-2+deb13u3 [1,806 kB]

Get:5 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf python3.13 armhf 3.13.5-2+deb13u3 [757 kB]

Get:6 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf libpython3.13-stdlib armhf 3.13.5-2+deb13u3 [1,815 kB]

Get:7 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf python3.13-minimal armhf 3.13.5-2+deb13u3 [1,802 kB]

Get:8 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf libpython3.13-minimal armhf 3.13.5-2+deb13u3 [848 kB]

Get:10 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf libexpat1-dev armhf 2.8.3-1~deb13u1 [153 kB]

Get:9 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf javascript-common all 12+nmul [4,864 B]

Get:11 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf libjs-jquery all 3.6.1+dfsg+~3.5.14-1 [326 kB]

Get:13 <http://mirror.netzwerke.de/raspbian/raspbian> trixie/main armhf libjs-sphinxdoc all 8.1.3-5 [30.5 kB]

Get:12 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf libjs-underscore all 1.13.4~dfsg+~1.11.4-3 [116 kB]

Get:15 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf libpython3.13-dev armhf 3.13.5-2+deb13u3 [3,808 kB]

Get:14 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf zlib1g-dev armhf 1:1.3.dfsg+really1.3.1-1 [903 kB]

Get:16 <http://mirror.netzwerke.de/raspbian/raspbian> trixie/main armhf libpython3-dev armhf 3.13.5-1 [10.5 kB]

Get:17 <http://raspbian.raspberrypi.com/raspbian> trixie/main armhf python3.13-dev armhf 3.13.5-2+deb13u3 [505 kB]

Get:18 <http://mirror.netzwerke.de/raspbian/raspbian> trixie/main armhf python3-dev armhf 3.13.5-1 [26.1 kB]

Get:19 <http://mirror.netzwerke.de/raspbian/raspbian> trixie/main armhf python3-packaging all 25.0-1 [56.6 kB]

Get:20 <http://mirror.netzwerke.de/raspbian/raspbian> trixie/main armhf python3-wheel all 0.46.1-2 [21.7 kB]

Fetchd 15.4 MB in 7s (2,106 kB/s)

apt-listchanges: Reading changelogs...

(Reading database ... 67462 files and directories currently installed.)

Preparing to unpack .../00-libexpat1_2.8.3-1~deb13u1_armhf.deb ...
Unpacking libexpat1:armhf (2.8.3-1~deb13u1) over (2.7.1-2) ...
Preparing to unpack .../01-python3.13-venv_3.13.5-2+deb13u3_armhf.deb ...
Unpacking python3.13-venv (3.13.5-2+deb13u3) over (3.13.5-2+deb13u2) ...
Preparing to unpack .../02-libpython3.13_3.13.5-2+deb13u3_armhf.deb ...
Unpacking libpython3.13:armhf (3.13.5-2+deb13u3) over (3.13.5-2+deb13u2) ...
Preparing to unpack .../03-python3.13_3.13.5-2+deb13u3_armhf.deb ...
Unpacking python3.13 (3.13.5-2+deb13u3) over (3.13.5-2+deb13u2) ...
Preparing to unpack .../04-libpython3.13-stdlib_3.13.5-2+deb13u3_armhf.deb ...
Unpacking libpython3.13-stdlib:armhf (3.13.5-2+deb13u3) over (3.13.5-2+deb13u2) ...
Preparing to unpack .../05-python3.13-minimal_3.13.5-2+deb13u3_armhf.deb ...
Unpacking python3.13-minimal (3.13.5-2+deb13u3) over (3.13.5-2+deb13u2) ...
Preparing to unpack .../06-libpython3.13-minimal_3.13.5-2+deb13u3_armhf.deb ...
Unpacking libpython3.13-minimal:armhf (3.13.5-2+deb13u3) over (3.13.5-2+deb13u2) ...
Selecting previously unselected package javascript-common.
Preparing to unpack .../07-javascript-common_12+nmul_all.deb ...
Unpacking javascript-common (12+nmul) ...
Selecting previously unselected package libexpat1-dev:armhf.
Preparing to unpack .../08-libexpat1-dev_2.8.3-1~deb13u1_armhf.deb ...
Unpacking libexpat1-dev:armhf (2.8.3-1~deb13u1) ...
Selecting previously unselected package libjs-jquery.
Preparing to unpack .../09-libjs-jquery_3.6.1+dfsg+~3.5.14-1_all.deb ...
Unpacking libjs-jquery (3.6.1+dfsg+~3.5.14-1) ...
Selecting previously unselected package libjs-underscore.
Preparing to unpack .../10-libjs-underscore_1.13.4+dfsg+~1.11.4-3_all.deb ...
Unpacking libjs-underscore (1.13.4+dfsg+~1.11.4-3) ...
Selecting previously unselected package libjs-sphinxdoc.
Preparing to unpack .../11-libjs-sphinxdoc_8.1.3-5_all.deb ...
Unpacking libjs-sphinxdoc (8.1.3-5) ...
Selecting previously unselected package zlib1g-dev:armhf.
Preparing to unpack .../12-zlib1g-dev_1%3a1.3.dfsg+really1.3.1-1_armhf.deb ...
Unpacking zlib1g-dev:armhf (1:1.3.dfsg+really1.3.1-1) ...
Selecting previously unselected package libpython3.13-dev:armhf.
Preparing to unpack .../13-libpython3.13-dev_3.13.5-2+deb13u3_armhf.deb ...
Unpacking libpython3.13-dev:armhf (3.13.5-2+deb13u3) ...
Selecting previously unselected package libpython3-dev:armhf.

Preparing to unpack .../14-libpython3-dev_3.13.5-1_armhf.deb ...
Unpacking libpython3-dev:armhf (3.13.5-1) ...
Selecting previously unselected package python3.13-dev.
Preparing to unpack .../15-python3.13-dev_3.13.5-2+deb13u3_armhf.deb ...
Unpacking python3.13-dev (3.13.5-2+deb13u3) ...
Selecting previously unselected package python3-dev.
Preparing to unpack .../16-python3-dev_3.13.5-1_armhf.deb ...
Unpacking python3-dev (3.13.5-1) ...
Selecting previously unselected package python3-packaging.
Preparing to unpack .../17-python3-packaging_25.0-1_all.deb ...
Unpacking python3-packaging (25.0-1) ...
Selecting previously unselected package python3-wheel.
Preparing to unpack .../18-python3-wheel_0.46.1-2_all.deb ...
Unpacking python3-wheel (0.46.1-2) ...
Selecting previously unselected package python3-pip.
Preparing to unpack .../19-python3-pip_25.1.1+dfsg-1+rpt1_all.deb ...
Unpacking python3-pip (25.1.1+dfsg-1+rpt1) ...
Setting up libexpat1:armhf (2.8.3-1~deb13u1) ...
Setting up javascript-common (12+nmul) ...
Setting up libpython3.13-minimal:armhf (3.13.5-2+deb13u3) ...
Setting up python3-packaging (25.0-1) ...
Setting up libexpat1-dev:armhf (2.8.3-1~deb13u1) ...
Setting up zlib1g-dev:armhf (1:1.3.dfsg+really1.3.1-1) ...
Setting up python3.13-minimal (3.13.5-2+deb13u3) ...
Setting up libjs-jquery (3.6.1+dfsg+~3.5.14-1) ...
Setting up libpython3.13-stdlib:armhf (3.13.5-2+deb13u3) ...
Setting up libjs-underscore (1.13.4~dfsg+~1.11.4-3) ...
Setting up libpython3.13:armhf (3.13.5-2+deb13u3) ...
Setting up python3.13 (3.13.5-2+deb13u3) ...
Setting up python3.13-venv (3.13.5-2+deb13u3) ...
Setting up python3-wheel (0.46.1-2) ...
Setting up libpython3.13-dev:armhf (3.13.5-2+deb13u3) ...
Setting up python3-pip (25.1.1+dfsg-1+rpt1) ...
Setting up libjs-sphinxdoc (8.1.3-5) ...
Setting up libpython3-dev:armhf (3.13.5-1) ...
Setting up python3.13-dev (3.13.5-2+deb13u3) ...
Setting up python3-dev (3.13.5-1) ...
Processing triggers for systemd (257.9-1~deb13u1+rpil) ...

```

Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+rpt1+deb13u3) ...
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Requirement already satisfied: pip in /opt/devialet/venv/lib/python3.13/site-
packages (25.1.1)
Collecting pip
  Downloading https://www.piwheels.org/simple/pip/pip-26.2.1-py3-none-
any.whl.metadata (4.6 kB)
Downloading https://www.piwheels.org/simple/pip/pip-26.2.1-py3-none-any.whl
(1.8 MB)
_____ 1.8/1.8
MB 1.8 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.1.1
    Uninstalling pip-25.1.1:
      Successfully uninstalled pip-25.1.1
Successfully installed pip-26.2.1
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Collecting paho-mqtt==2.1.0 (from -r /opt/devialet/requirements.txt (line 1))
  Downloading paho_mqtt-2.1.0-py3-none-any.whl.metadata (23 kB)
Collecting pyserial==3.5 (from -r /opt/devialet/requirements.txt (line 2))
  Downloading pyserial-3.5-py2.py3-none-any.whl.metadata (1.6 kB)
Downloading paho_mqtt-2.1.0-py3-none-any.whl (67 kB)
Downloading pyserial-3.5-py2.py3-none-any.whl (90 kB)
Installing collected packages: pyserial, paho-mqtt
Successfully installed paho-mqtt-2.1.0 pyserial-3.5

PAS EERST /etc/devialet/devialet.env AAN EN START DAARNA DE SERVICE.

```

Note:

*The last line in **bold** is translated in English: "IMPORTANT: Configure /etc/devialet /devialet.env before starting the service."*

These are value from Home Assistant.

```

MQTT_HOST=192.168.1.10
MQTT_PORT=1883
MQTT_USERNAME=devialet
MQTT_PASSWORD=<replace-me>

```

Not setting the parameters will limit the functionality to only the web application.

8.8 Starting up the service

```

devialet@devialet-rs232:~/devialet-pi $ sudo systemctl start devialet-
bridge.service
[sudo] password for devialet:*****

```

8.9 Verify service is started

```
devialet@devialet-rs232:~/devialet-pi $ systemctl --no-pager --full status devialet-bridge.service
```

- devialet-bridge.service - Devialet RS-232, MQTT en web bridge

Loaded: loaded (/etc/systemd/system/devialet-bridge.service; disabled; preset: enabled)

Active: active (running) since Fri 2026-09-11 20:20:10 CEST; 10s ago

Invocation: 507497e5aa954336af09006c93af7b13

Main PID: 3777 (python)

Tasks: 3 (limit: 449)

CPU: 3.520s

CGroup: /system.slice/devialet-bridge.service

└─3777 /opt/devialet/venv/bin/python
/opt/devialet/src/devialet_bridge.py

8.10 Verify if the Web application is working

Check in local browser URL: <http://devialet-rs232.local:8080>

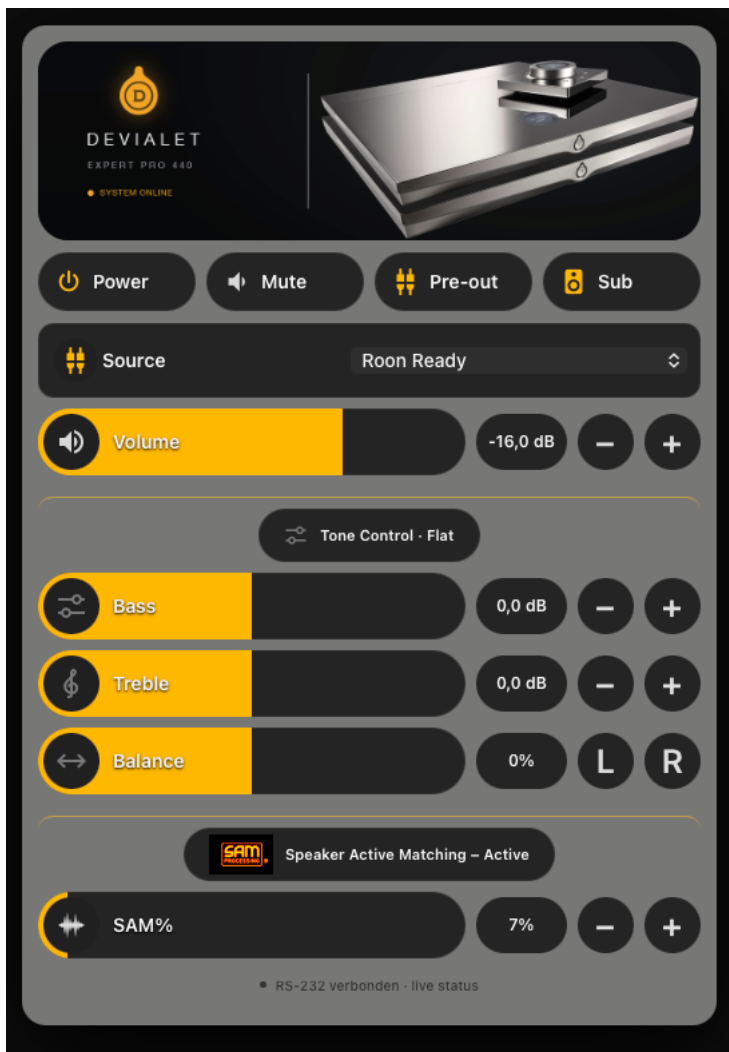


Figure 20. Devialet-rs232 Web App

8.11 Enable service at startup (boot)

Assuming the functional test was successful, enable the service to start at power on or reboot from the Raspberry Pi.

```
devialet@devialet-rs232:~/devialet-pi $ sudo systemctl enable devialet-bridge.service
```

```
[sudo] password for devialet: *****
```

```
Created symlink '/etc/systemd/system/multi-user.target.wants/devialet-bridge.service' → '/etc/systemd/system/devialet-bridge.service'.
```

8.12 Test a reboot

The service should start during reboot.

```
devialet@devialet-rs232:~/devialet-pi $ sudo reboot
```

Check in local browser URL: <http://devialet-rs232.local:8080>

9 Devialet RS-232 command reference

Based on the Devialet documentation and testing via Minicom on the Raspberry Pi the working control commands on a Devialet Expert Pro 220 / 440 are displayed in table 1.

Some controls are only RO (Read Only), to read a status.

The most useful controls are the controls that are also R/W (readable and writable), to change a setting.

All commands start with: **[Deviallet>** followed by **“CONTROL”=“value”]**

Examples:

-
- **[Deviallet>POWER=1]** will power on the Devialet
- **[Deviallet>SOURCE=?]** will display the current source
- **[Deviallet>VOLUME=-20.0]** will set the volume.
- **[Deviallet>CURVE=-1]** will display an error

If a parameter is not in the range, an error is returned: **INVALID PARAMETER**

Table 1. Devialet RS-232 controls and parameter ranges

Control	R/W or RO	Range	Steps	Comment
POWER	R/W	0/1		
START	RO	0/1		Busy starting
MUTE	R/W	0/1		
VOLUME	R/W	-97.5 / +20	0.5	
SOURCE	R/W	List		SOURCE LIST
BALANCE	R/W	-100 / +100	1	
TONE_CONTROL	RO	0/1		Set to 1 when BASS or TREBLE are not 0
BASS	R/W	-18 / +18	0.5	
TREBLE	R/W	-18 / +18	0.5	
MATCHING_LEVEL	R/W	0-100	1	SAM % (not SAN ON/OFF)
SUBSONIC_FILTER	R/W	0/1		
SUBWOOFER	R/W	0/1		
PREOUT	R/W	0/1		
PHASE	R/W	0/1		
CURVE	R/W	0 / 13		RIAA CURVE LIST
USER_DSP	RO	0/1		To be tested
VOLUME_BYPASS	RO	0/1		To be tested

10 Home Assistant integration

Add MQTT in HA

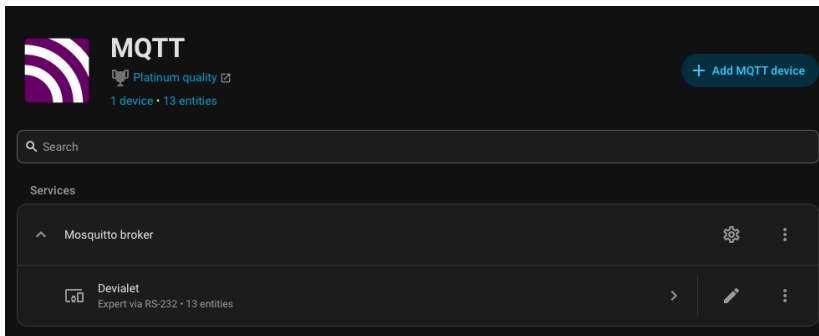


Figure 21. MQTT Installation step

Configure username / password in HA

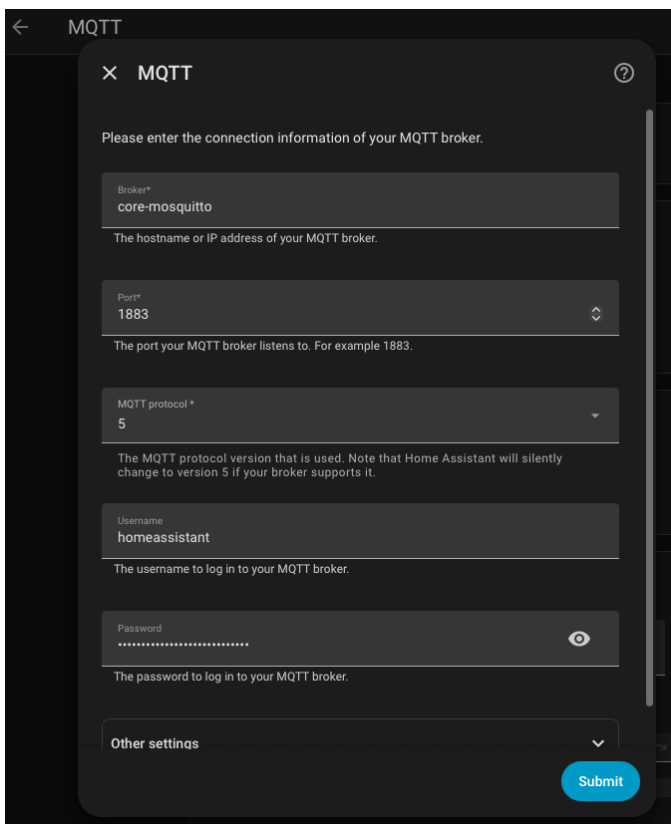


Figure 22. MQTT Installation step

Note: Set the same username & password in the .ENV file on the Pi.

When done, the Devialet controls are now visible in HA as Entity's.

10.1 Create a custom dashboard

11 Devialet Configurator settings

Remote settings Button 2 and 3 can be configured in the on-line Devialet Configurator to:

- *Disables*
- *RAM / Subsonic*
- *Phase invert* (thus Phase is eigenlijk “*Phase invert*”)
- *Subwoofer*
- *Preamplified Output*
- *Subsonic filter*
- *Balance & Tone control*
- *Mute*
- *SAM On/Off*

ICM = Intelligent Cinema Mode

